Junyoung Kim

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RESEARCH INTERESTS

My research focuses on how search and conversational systems can help people who can't quite say what they want, and how to evaluate them when no single answer is right.

1. **Resolving Underspecified Intent** [C1, C4]: Exploring multiple plausible interpretations when a query or request is incomplete or ambiguous, and narrowing them with signals such as what users dislike.
2. **Evaluation Beyond a Single Answer** [C3]: Reference-free evaluation of open-ended outputs where many different answers can be appropriate.

EDUCATION

Sungkyunkwan University Sep 2023 – Aug 2025
M.S. in Artificial Intelligence, (GPA: 4.36/4.5)
Advisor: Prof. Jongwuk Lee

University of Toronto Jan 2025 – Jun 2025
Graduate Visiting Student, Mechanical & Industrial Engineering, (GPA: 4.0/4.0)
Full fellowship from MSIT & IITP, Korea

Sungkyunkwan University Mar 2019 – Aug 2023
B.S. in Computer Education, (GPA: 4.17/4.5), *Magna Cum Laude*

PUBLICATIONS

**: Equal contribution.*

International Conferences

[C1] **Bayesian Active Learning with Gaussian Processes Guided by LLM Relevance Scoring for Dense Passage Retrieval** |
Junyoung Kim, Anton Korikov, Jiazhou Liang, Justin Cui, Yifan Simon Liu, Qianfeng Wen, Mark Zhao, Scott Sanner
Annual Meeting of the Association for Computational Linguistics (***ACL***), *Findings, 2026*

[C2] **Multimodal Item Scoring for Natural Language Recommendation via Gaussian Process Regression with LLM Relevance Judgments** |
Yifan Simon Liu, Qianfeng Wen, Jiazhou Liang, Mark Zhao, Justin Cui, Anton Korikov, Armin Toroghi, **Junyoung Kim**, Scott Sanner
Annual Meeting of the Association for Computational Linguistics (***ACL***), *Findings, 2026*

[C3] **Transparent Reference-free Automated Evaluation of Open-Ended User Survey Responses** |
Subin An*, Yugyeong Ji*, **Junyoung Kim***, Heejin Kook*, Yang Lu, Josh Seltzer
Conference on Empirical Methods in Natural Language Processing (***EMNLP***), *Industry Track, 2025*

[C4] **Empowering Retrieval-based Conversational Recommendation with Contrasting User Preferences** |
Heejin Kook*, **Junyoung Kim***, Seongmin Park, Jongwuk Lee
Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (***NAACL***), *2025*

[C5] **MARS: Matching Attribute-aware Representations for Text-based Sequential Recommendation** |
Hyunsoo Kim*, **Junyoung Kim***, Minjin Choi, Sunkyung Lee, Jongwuk Lee
ACM International Conference on Information and Knowledge Management (***CIKM***), *Short Paper, 2024*

Domestic Journals

[J1] **An Inference Framework for Text-Based Sequential Recommendation Model Using Nearest Neighbor Mechanism**
Junyoung Kim, Hyunsoo Kim, Jongwuk Lee
Journal of KIISE, 2025

[J2] **A Deep Analysis of the Unified Recommender Model Using Pre-trained Language Models and Suggestions for Improvement**
Junyoung Kim*, Hyunsoo Kim*, Jongwuk Lee
Journal of KIISE, 2024. Extended version of a *KSC 2022* paper [Best Presentation]

RESEARCH EXPERIENCE

Data-Driven Decision-Making (D3M) Lab, University of Toronto Jan 2025 – Aug 2025
Research Assistant, advised by Prof. Scott Sanner

- Led research on cost-aware document relevance modeling under LLM inference constraints, improving Recall@50 by up to 42.2% over existing methods [C1].
- Contributed to research on LLM-based relevance modeling for natural-language recommendation [C2].

Nexxt Intelligence Jan 2025 – Jul 2025
Student Researcher (Industry-Academia Collaboration, Ministry of Science and ICT)

- Developed an LLM-as-a-Judge framework for evaluating open-ended survey responses, achieving up to 0.86 Spearman correlation with human experts [C3].

Data Intelligence and Learning (DIAL) Lab, Sungkyunkwan University Apr 2022 – Jan 2025
Research Assistant, advised by Prof. Jongwuk Lee

- Led research on LLM-based conversational recommendation with contrastive user preference modeling, improving Recall@10 by up to 99.7% over existing models [C4].
- Led research on attribute-aware text-based sequential recommendation, improving NDCG@10 by up to 29.3% (13%+ in zero-shot settings) over existing methods [C5].

INDUSTRY EXPERIENCE

Naver Corp. Jul 2026 – Oct 2026
AI Engineering Intern

- Developed a self-improving LLM-as-a-Judge system to automate relevance annotation of search data, reducing manual labeling.
- Developed an LLM-based system that translates natural-language label modification requests into executable code, letting annotators revise labels without programming.

Angle Company Jan 2026 – Mar 2026
AI Engineering Intern

- Developed handwriting separation and preprocessing methods for OCR on insurance documents.

AWARDS

***2nd Prize* (*Stage 2, 9 teams*) *& 4th Prize* (*Stage 1, 54 teams*), *Ministerial Award*, 6th AI Grand Challenge** 2022, 2023
Developed a multimodal RAG system for policy decision support

***3rd Prize*, 4th AI x BOOKATHON (AI Writing Contest)** 2023
Keyword-driven essay generator that maintains a consistent plot across long context

***Best Presentation Award*, Korea Software Congress** 2022

***1st Prize*, SKKU AI-Education Hackathon** 2022
Visual analytics system to assist peer learning based on source code

***2nd Prize*, SKKU AI-Education Hackathon** 2021
Korean narrative quiz generator based on word embedding

FELLOWSHIPS & SCHOLARSHIPS

Full Fellowship, University of Toronto AI Convergence Program, MSIT & IITP, Korea 2025
SKKU AI Graduate Student Research Grant 2023, 2024
SKKU Academic Excellence Scholarship 2022, 2023

TEACHING EXPERIENCE

Teaching Assistant
Database Project 2024 Fall
Introduction to Database 2024 Spring
Fundamentals of Machine Learning 2023 Fall